Review Article

Biologics in Chronic Rhinosinusitis with Nasal Polyposis: Current Evidence, Treatment Protocols and Long-Term Safety Considerations

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Abstract

Chronic rhinosinusitis with nasal polyposis (CRSwNP) is a persistent inflammatory disorder of the upper airway, frequently associated with type 2 inflammation, asthma, allergic disease and aspirin/non-steroidal anti-inflammatory drug-exacerbated respiratory disease. Conventional treatment with saline irrigation, intranasal corticosteroids, intermittent systemic corticosteroids and endoscopic sinus surgery remains effective for many patients; however, a subgroup develops recurrent, severe disease with persistent nasal obstruction, anosmia, impaired quality of life and repeated exposure to systemic corticosteroids or revision surgery. Biologic therapies have changed the management landscape by targeting key immunologic pathways such as IL-4/IL-13 signaling, IgE, IL-5, IL-5 receptor alpha and epithelial alarmins such as thymic stromal lymphopoietin. This narrative review expands a previously prepared editorial into a review-style manuscript, summarizing the rationale for biologic therapy, candidate selection, practical treatment protocols, outcome monitoring, long-term safety considerations, steroid-sparing benefits, economic barriers and future directions. Current evidence supports biologics as add-on maintenance therapy in carefully selected patients with severe uncontrolled CRSwNP, particularly where type 2 inflammation and comorbid lower-airway disease are evident. Objective reassessment using nasal polyp score, symptom scores, SNOT-22, smell testing, systemic corticosteroid need and revision surgery requirement is essential to guide continuation, switching or discontinuation. Long-term data are reassuring but ongoing pharmacovigilance, registry-based follow-up, cost-effectiveness studies and biomarker-driven selection are needed to refine personalized care.

Keywords: chronic rhinosinusitis with nasal polyps; nasal polyposis; biologics; dupilumab; omalizumab; mepoli zumab; tezepelumab; type 2 inflammation; endoscopic sinus surgery; steroid-sparing therapy.

Introduction:

Chronic rhinosinusitis with nasal polyposis (CRSwNP) is a chronic inflammatory disease characterized by persistent sinonasal inflammation and the formation of benign inflammatory polyps. The disease is clinically important because it can produce long-standing nasal obstruction, hyposmia or anosmia, facial pressure, rhinorrhea, sleep disturbance, recurrent infective exacerbations and substantial impairment of health-related quality of life. Although intranasal corticosteroids and endoscopic sinus surgery have remained central to management, disease recurrence after surgery and the adverse effects of repeated systemic corticosteroids have created a need for targeted, disease-modifying therapy [1,2].

The past decade has seen rapid progress in understanding the immunologic heterogeneity of CRSwNP. In many patients, particularly in Western and increasingly in global cohorts, the dominant inflammatory pattern is type 2 inflammation, driven by interleukin (IL)-4, IL-5, IL-13, IgE, eosinophils, mast cells and epithelial-derived cytokines. This endotype overlaps with asthma, allergic rhinitis and aspirin-exacerbated respiratory disease, supporting the concept of a unified airway in which upper- and lower-airway inflammation should be managed together [2]. Biologic therapies exploit this pathobiology by interrupting specific inflammatory pathways rather than merely reducing mucosal edema or removing polyps surgically.

2. Review Methodology:

This narrative review was prepared by expanding the supplied editorial draft into a review-article format and updating the discussion with recent position papers, pivotal phase 3 trials, prescribing information and real-world systematic reviews. The emphasis is practical rather than systematic: patient selection, initiation protocols, monitoring, safety and implementation issues are prioritized for clinicians managing severe uncontrolled CRSwNP.

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3. Pathobiology and Rationale for Targeted Therapy:

CRSwNP is not a single disease entity. It is a clinical phenotype with several inflammatory endotypes. Type 2 inflammation is the best characterized and is usually associated with eosinophilic tissue inflammation, local IgE production, epithelial barrier dysfunction and increased expression of IL-4, IL-5 and IL-13. IL-4 and IL-13 promote IgE class switching, mucus production, epithelial remodeling and chemokine expression; IL-5 supports eosinophil maturation, recruitment and survival; IgE amplifies mast-cell and basophil activation; and thymic stromal lymphopoietin (TSLP) acts upstream by activating dendritic cells and innate lymphoid cells, thereby amplifying type 2 responses [2].

This mechanistic framework explains the current biologic classes used for CRSwNP. Anti-IL-4 receptor alpha therapy blocks shared IL-4 and IL-13 signaling; anti-IgE therapy reduces free IgE and downstream effector-cell activation; anti-IL-5 and anti-IL-5 receptor approaches reduce eosinophilic inflammation; and anti-TSLP therapy targets an epithelial alarmin that sits upstream of several inflammatory cascades. Because these pathways are also relevant to asthma and atopic disease, biologic selection should consider the whole airway and not only the endoscopic appearance of the nose.

Table 1. Major biologic options and their practical relevance.

Biologic	Target	Mechanism	Typical approved CRSwNP use/ dose*	Clinical relevance
Dupilumab	IL-4 receptor alpha	Blocks IL-4 and IL-13 signaling	US: adults and pediatric patients >=12 years; 300 mg SC every 2 weeks [3]	Broad type 2 inflammation; strong evidence for polyp burden, congestion, smell and quality-of-life improvement.
Omalizumab	IgE	Binds free IgE and reduces IgE-mediated effector-cell activation	US: adults with CRSwNP; 75-600 mg SC every 2 or 4 weeks based on IgE and body weight [4]	Useful where allergic phenotype, high IgE or comorbid allergic asthma is prominent.
Mepolizumab	IL-5	Reduces eosinophil survival and activation	US: adults with inadequate response to nasal corticosteroids; 100 mg SC every 4 weeks [5]	Useful in eosinophilic disease and recurrent, refractory CRSwNP.
Tezepelumab	TSLP	Blocks upstream epithelial alarmin signaling	US: >=12 years with inadequately controlled CRSwNP; 210 mg SC every 4 weeks [6]	Expands targeting beyond downstream type 2 cytokines; supported by WAYPOINT phase 3 data [10].
Benralizumab	IL-5 receptor alpha	Induces eosinophil depletion through antibody-dependent cell-mediated cytotoxicity	Approval status for CRSwNP varies; established in eosinophilic asthma	Investigational or selected-region option for eosinophilic CRSwNP; local approval must be checked.

*Examples reflect current US labeling where cited; local indications, reimbursement rules and age cut-offs may differ.

4. Current Biologic Options:

Regulatory approvals vary across countries, age groups and time. Table 1 summarizes major biologic options and their practical relevance. Clinicians should verify the most recent local prescribing information before initiating therapy.

5. Patient Selection: Who Should Receive a Biologic?

Biologics should not be viewed as first-line therapy for all patients with nasal polyps. They are most appropriate for severe uncontrolled CRSwNP despite optimized conventional care. Candidate selection should integrate disease severity, prior treatment burden, comorbidities, biomarkers and patient preference. Contemporary EPOS/EUFOREA guidance emphasizes type 2 inflammation, impact on quality of life, smell dysfunction, systemic corticosteroid exposure, need for surgery and lower-airway comorbidity [2].

Practical indications for considering biologic therapy include:

- Bilateral nasal polyposis with persistent symptoms de spite regular intranasal corticosteroids and saline irri gation.
- History of prior endoscopic sinus surgery with recur rent polyps, or a patient in whom surgery is contrain dicated, declined or unlikely to provide durable control.
- Repeated or prolonged systemic corticosteroid require ment, especially when steroid-related morbidity is pres ent or anticipated.
- High symptom burden, severe smell loss, high SNOT-22 score or marked impairment in sleep and daily func tioning.
- Evidence of type 2 inflammation, such as tissue eo sin ophilia, blood eosinophilia, elevated IgE or comor bid asthma/AERD.
- Coexisting moderate-to-severe asthma where a single biologic may improve both upper- and lower-airway disease.

Shared decision-making is essential. The discussion should address expected benefits, onset of response, injection schedule, need for ongoing nasal corticosteroids, possible adverse events, cost, insurance or institutional access, and the uncertainty regarding ideal treatment duration.

6. Practical Protocol for Initiation and Moni toring:

Historically, biologics were often reserved for patients after multiple surgeries. Current practice is shifting toward earlier use in carefully selected patients to prevent recurrent systemic corticosteroid exposure and repeated procedures. A structured protocol improves consistency and helps avoid both under-treatment and unnecessary long-term expense.

6.1 Baseline Assessment:

Before initiation, document the diagnosis and severity. Nasal endoscopy should record bilateral nasal polyp score; CT findings should be reviewed when surgery is being considered or when the diagnosis is uncertain. Baseline patient-reported outcomes should include SNOT-22 and a visual analog scale for nasal obstruction, rhinorrhea, facial pressure and smell. Objective smell testing is desirable where available. Baseline systemic corticosteroid use, antibiotic courses, emergency visits, revision surgery history and asthma status should be recorded.

6.2 Multidisciplinary Review:

Collaboration among otorhinolaryngologists, allergists, pulmonologists and primary care physicians is increasingly important, especially for patients with asthma, AERD, allergic disease, eosinophilic granulomatosis with polyangiitis mimics, immunodeficiency or recurrent infections. Multidisciplinary assessment can also identify patients in whom biologic choice should be driven primarily by lower-airway disease or by a broader type 2 inflammatory phenotype.

6.3 Response Assessment:

Response should be assessed at predefined intervals, commonly at 4 to 6 months and again at 12 months, using the same measures obtained at baseline. A complete response is suggested by meaningful reduction in nasal polyp score, improved smell, reduced nasal congestion, reduced systemic corticosteroid need, improved SNOT-22 and improved control of comorbid asthma. Partial response may justify continuation with closer monitoring, optimization of topical therapy or consideration of switching. Non-response should prompt reassessment of diagnosis, adherence, comorbidities and local inflammatory phenotype.

Table 2. Suggested practical monitoring framework for biologic therapy in CRSwNP.

Domain	Baseline	Follow-up indicators
Endoscopic disease	Bilateral nasal polyp score; mucosal edema; discharge	Reduction in polyp size and inflammation
Symptoms	Nasal obstruction, rhinorrhea, facial pressure, sleep	Visual analog scale and symptom diary improvement
Quality of life	SNOT-22 or equivalent	Clinically meaningful reduction in score
Olfaction	Patient report; smell identification test where available	Improved smell identification or patient-reported smell
Treatment burden	OCS courses, antibiotics, prior surgery	Reduced need for OCS and revision surgery
Comorbid asthma	Asthma control, exacerbations, spirometry if relevant	Improved asthma control and reduced exacerbations

7. Evidence from Pivotal Trials and Real-World Studies

Dupilumab demonstrated significant improvements in endoscopic polyp score, sinus opacification, nasal congestion, smell and health-related quality of life in the LIBERTY NP SINUS-24 and SINUS-52 phase 3 trials and is considered one of the best-established options for severe CRSwNP [7]. Omalizumab improved endoscopic, clinical and patient-reported outcomes in the replicate POLYP 1 and POLYP 2 phase 3 trials, supporting IgE blockade as a useful option in severe disease with inadequate response to standard therapy [8]. Mepolizumab, evaluated in the SYNAPSE trial, improved nasal polyp size and nasal obstruction and reduced the need for surgery in patients with recurrent, refractory severe bilateral CRSwNP [9].

Tezepelumab has recently added a new upstream therapeutic approach. In the WAYPOINT phase 3 trial, tezepelumab significantly reduced nasal polyp size, nasal congestion severity and sinonasal symptom burden in severe uncontrolled CRSwNP [10]. This is clinically relevant because epithelial alarmins may drive inflammation across a broader spectrum of patients than those defined by a single downstream biomarker. Real-world studies and meta-analyses are broadly reassuring regarding effectiveness and safety, but they also highlight heterogeneity in response, the need for standard response definitions and the importance of long-term follow-up [11,12].

8. Long-Term Safety Considerations:

The long-term safety profile of biologics in CRSwNP is generally reassuring, particularly when compared with the cumulative harms of repeated systemic corticosteroid exposure. Nonetheless, biologics are chronic immune-modulating therapies and require ongoing vigilance. Common adverse events include injection-site reactions, upper respiratory tract infections, nasopharyngitis, headache, arthralgia, transient eosinophilia and drug-specific events such as conjunctivitis with dupilumab in some patients. Hypersensitivity reactions are uncommon but clinically important. Helminth infection considerations and live-vaccine cautions appear in several biologic labels and should be reviewed before initiation [3-6].

No consistent signal of increased malignancy, opportunistic infection or severe immunosuppression has emerged from available trial and real-world data, but registry-based surveillance remains essential because many patients require prolonged therapy. Clinicians should avoid abrupt withdrawal of systemic or inhaled corticosteroids when biologics are started; tapering should be gradual and clinically supervised. Special populations such as pregnancy, lactation, elderly patients, patients with immunodeficiency and pediatric patients require individualized risk-benefit assessment based on local labeling and specialty consultation.

Table 3. Safety and follow-up considerations during biologic therapy.

Issue	Examples	Practical approach
Common adverse events	Injection-site reactions, nasopharyngitis, upper respiratory tract infection, headache, arthralgia	Explain before initiation; record at each visit.
Drug-specific cautions	Conjunctivitis/keratitis reported with dupilumab; herpes zoster warning for mepolizumab; hypersensitivity warnings across agents	Review label-specific warnings and past history.
Steroid tapering	Risk of adrenal suppression or disease flare if corticosteroids are stopped abruptly	Taper gradually and monitor asthma and sinonasal disease.
Infection/vaccination	Helminth infection precautions; avoidance of live attenuated vaccines in some labels	Screen history and update vaccination plan where appropriate.
Long-term uncertainty	Optimal duration, stopping rules and switching strategy remain incompletely defined	Use response criteria, shared decision-making and registry follow-up.

9. Steroid-Sparing and Surgery-Sparing Value:

Repeated systemic corticosteroid exposure is associated with diabetes, hypertension, osteoporosis, cataract, weight gain, mood disturbance, adrenal suppression and increased infection risk. A major advantage of biologic therapy is the potential to reduce corticosteroid dependence while improving symptoms and quality of life. In selected patients, biologics may also delay or reduce revision surgery. This does not eliminate the role of surgery: endoscopic sinus surgery remains important for mechanical obstruction, tissue diagnosis, restoration of sinus ventilation, access for topical therapy and rapid debulking in appropriate patients. The emerging paradigm is not biologics versus surgery, but individualized sequencing and combination of surgery, topical therapy and biologic treatment.

10. Economic and Access Challenges:

Despite their clinical promise, biologics create substantial economic and logistical challenges. High acquisition costs, injection training, cold-chain storage, reimbursement variability and need for prolonged administration can limit access, particularly in low-resource settings. Economic models often compare biologics with revision surgery, but the most relevant comparison differs by patient: some patients primarily need steroid-sparing control; others need surgery to restore anatomy; still others require a biologic because of severe comorbid asthma. Cost-effectiveness research should incorporate quality-adjusted life years, corticosteroid morbidity, surgery avoidance, asthma benefits, adherence and patient preference.

11. Switching, Discontinuation and Duration of Therapy:

The ideal duration of biologic therapy remains one of the major unanswered questions. Most pivotal trials provide data up to approximately one year, while real-world cohorts are beginning to describe longer-term outcomes. Some patients relapse when treatment is stopped, suggesting that ongoing immune modulation may be required for disease control. For partial responders, switching to another biologic with a different mechanism is reasonable after reassessment of adherence, topical therapy, comorbid asthma, aspirin sensitivity and diagnostic accuracy. Discontinuation may be considered in non-responders, patients with unacceptable adverse effects, patients with sustained remission who prefer a monitored trial off treatment, or when cost or access becomes prohibitive. Any discontinuation strategy should include a clear follow-up plan and rescue pathway.

12. Future Directions:

Future progress will depend on precision medicine. Current selection criteria rely on clinical severity, history of systemic corticosteroid use, prior surgery and broad type 2 markers. More accurate biomarkers are needed to predict which patient will respond best to anti-IL-4/IL-13, anti-IgE, anti-IL-5, anti-IL-5 receptor or anti-TSLP therapy. Molecular endotyping, transcriptomic profiling, local tissue biomarkers, peripheral eosinophil dynamics and integrated upper-lower airway phenotyping may eventually support more rational biologic choice. Other research priorities include optimal timing relative to surgery, standardized definitions of remission, pediatric outcomes, pregnancy safety, low-resource implementation and cost-effective treatment algorithms.

13. Conclusion:

Biologics have ushered in a new era in the management of severe uncontrolled CRSwNP. By targeting the immune pathways responsible for type 2 and epithelial-driven inflammation, these therapies can reduce polyp burden, improve nasal obstruction and smell, enhance quality of life and reduce dependence on systemic corticosteroids and revision surgery in appropriately selected patients. Successful implementation requires careful candidate selection, multidisciplinary assessment, objective monitoring and regular review of response, safety and cost. Surgery and topical corticosteroid therapy remain essential parts of care, but biologics have redefined the standard for patients with recurrent, severe and steroid-dependent disease. The next phase of progress will be driven by biomarkers, long-term registry data and precision protocols that match the right biologic to the right patient at the right time.

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Conflicts of interest: None

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